Chiranth K

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EDUCATION

The National Institute of Engineering — 2022 – Present

*B.E. Information Science Engineering – **CGPA: 8.6/10.0*** — *Mysuru, India*

EXPERIENCE

Research Intern – VISTA Lab (Advisor: Dr. Punit Rathore) — Mar 2026 – Present

Indian Institute of Science (IISc) — *Bangalore, India*

- Analyzed online and offline multi-object tracking methods for traffic surveillance, implementing and evaluating multiple tracking algorithms to understand identity preservation, track association, and failure modes in real-world CCTV data.
- Contributing to a new dataset and comprehensive benchmark paper targeting **CVPR 2027**, advancing multi-object tracking research in urban traffic scenarios.
- **Certificate:** Ongoing

Research Intern – Deep Learning and Computer Vision Lab (Advisor: Dr. Shyam Lal) — July 2025 – February 2026

National Institute of Technology Karnataka (NITK) — *Surathkal, India*

- Conducted dual-domain deep learning research in Remote Sensing and Medical Imaging.
- Developed a controlled benchmarking framework for high-resolution building change detection on LEVIR-CD and EGY-BCD datasets, managing preprocessing for 6,000+ image pairs.
- Evaluated 8 widely used change detection architectures under a standardized evaluation framework to build consistent performance baselines.
- **Certificate:** View

Deep Learning Intern — Apr 2025 – May 2025

AIQuantum Smart Solutions Pvt. Ltd., NITK STEP Incubator — *Remote*

- Implemented attention-augmented U-Net variants for road segmentation on aerial satellite imagery.
- Reimplemented convolution, pooling, and activation layers from scratch in NumPy to validate gradient propagation and numerical stability against framework outputs.
- **Certificate:** View

PUBLICATIONS

In Preparation

- Sujan D*, **Chiranth Kumaraswamy***, Sravya N, Vibha Damodara Kevala, Shyam Lal, Shilpa Suresh, Jayasri B.S., Bhat Geetalaxmi Jairam. *"EBCDNet - Enhanced Building Change Detection Network from High Resolution Remote Sensing Images."*

*Co-first authors (equal contribution).

PROJECTS

Histopathological Cancer Classification (Colon & Prostate) | Pytorch, Keras, NumPy, OpenCV | GitHub — **2026**

- Reproduced and benchmarked multiple state-of-the-art lightweight architectures for histopathology classification across colon and prostate cancer datasets.
- Conducted controlled cross-architecture evaluation using accuracy, F1-score, precision, recall, and generalization gap analysis.
- Analyzed contextual channel attention and gating mechanisms underlying existing histopathology networks.
- Designing a lightweight architecture targeting improved efficiency and generalization over existing models.

Bi-Temporal Change Detection Research Framework | TensorFlow, NumPy | GitHub — **2025**

- Engineered **EBCDNet**, achieving $\sim$1.5–2% IoU improvement over existing models while maintaining faster inference speeds through optimized feature fusion.
- Conducted quantitative error analysis on extreme class imbalance ($<$ 5% foreground pixels), identifying and mitigating failure modes in boundary localization.
- Developed modular experimental scripts to validate model generalization across regional benchmarks, evaluating performance on the LEVIR-CD (U.S.) and EGY-BCD (Egypt) datasets.

Road Segmentation using Deep Learning | TensorFlow, OpenCV | GitHub **2025**

- Implemented attention-enhanced U-Net for aerial road segmentation with modular loss and augmentation strategies.
- Improved IoU and F1-score over baseline U-Net through controlled architectural experimentation.

PLANTEFY: Plant Species Recognition | TensorFlow, OpenCV, Android | GitHub **2024**

- Designed sequential and skip-connected CNN architectures for plant species classification.
- Optimized preprocessing and deployed a lightweight Android application for real-time inference.

INTERESTS

- Machine Learning and Deep Learning
- Quantum Computing
- Recreational mathematics (Euclidean geometry)